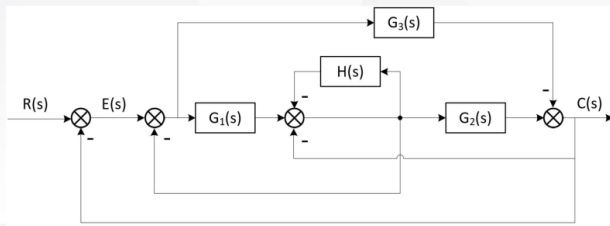


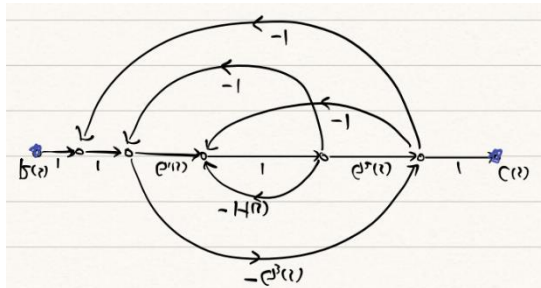
控制系统的框图如下所示:

(1) 使用梅森增益公式求传递函数 $C(s)/R(s)$;

(2) 使用梅森增益公式求传递函数 $E(s)/R(s)$



转化成信号流图如下:



前向通道:

$$P_1 = G_1(s)G_2(s)$$

$$P_2 = -G_3(s)$$

回路:

$$L_1 = -G_1(s), \quad L_2 = -G_2(s), \quad L_3 = -H(s), \quad L_4 = -G_3(s)$$

$$L_5 = -G_1(s)G_2(s), \quad L_6 = G_3(s)$$

不接触回路

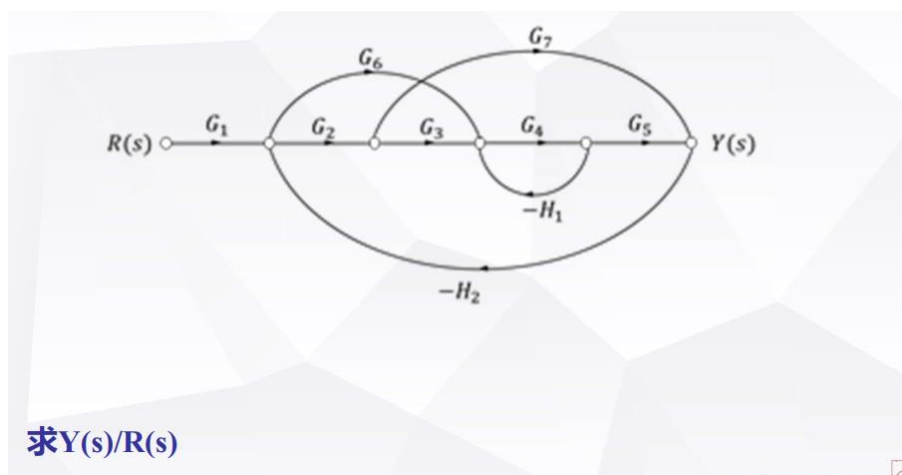
$$L_3L_6 = -H(s)G_3(s)$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 - L_3$$

$$\frac{C}{R} = \frac{G_1G_2 - G_3(1+H)}{1 + H + G_1 + G_2 - G_3H + G_1G_2}$$

$$\frac{E}{R} = \frac{H + G_1 + G_2 + G_3 + H}{1 + H + G_1 + G_2 - G_3H + G_1G_2}$$



求 $Y(s)/R(s)$

前向通路：

$$P_1 = G_1 G_2 G_3 G_4 G_5, \quad P_2 = G_1 G_6 G_4 G_3, \quad P_3 = G_1 G_2 G_7$$

回路：

$$L_1 = -H_1 G_4, \quad L_2 = -H_2 G_2 G_3 G_4 G_5, \quad L_3 = -H_2 G_6 G_4 G_5,$$

$$L_4 = -H_2 G_2 G_7,$$

不接触回路： $L_4 L_1 = -H_1 H_2 G_2 G_4 G_7$,

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + L_1 L_4$$

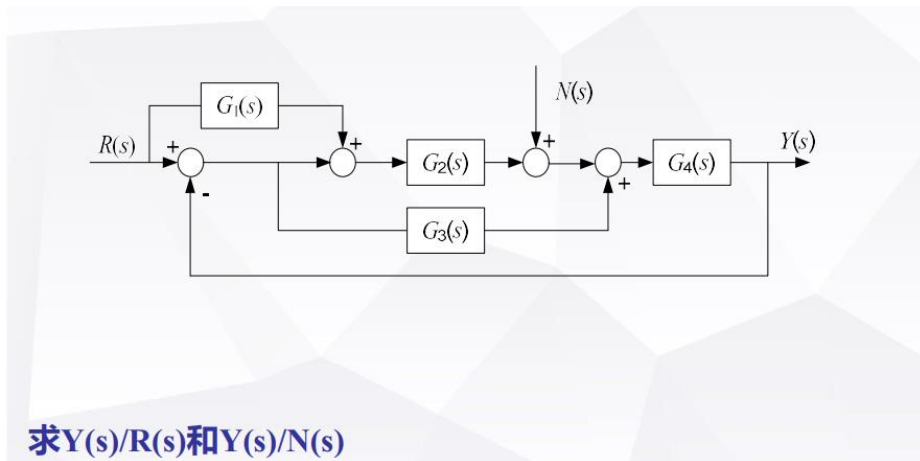
$$\Delta_1 = 1$$

$$\Delta_2 = 1$$

$$\Delta_3 = 1 - L_1 = 1 + H_1 G_4$$

$$\frac{Y(S)}{R(S)} = \frac{1}{\Delta} (p_1 \Delta_1 + p_2 \Delta_2 + p_3 \Delta_3)$$

$$= \frac{G_1 G_2 G_3 G_4 G_5 + G_1 G_4 G_5 G_6 + G_1 G_2 G_7 (1 + H_1 G_4)}{1 + H_1 G_4 + H_2 G_2 G_3 G_4 G_5 + H_2 G_4 G_5 G_6 + H_2 G_2 G_7 + H_1 H_2 G_2 G_4 G_7}$$



前向通道： $P_1 = G_1(s)G_2(s)G_4(s)$, $P_2 = G_2(s)G_4(s)$, $P_3 = G_3(s)G_4(s)$

单独回路： $L_1 = -G_2(s)G_4(s)$, $L_2 = -G_3(s)G_4(s)$

特征式： $\Delta = 1 + G_2(s)G_4(s) + G_3(s)G_4(s)$, $\Delta_1 = \Delta_2 = \Delta_3 = 1$

$$\text{则 } Y(s)/R(s) = \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3) = \frac{G_1(s)G_2(s)G_4(s) + G_2(s)G_4(s) + G_3(s)G_4(s)}{1 + G_2(s)G_4(s) + G_3(s)G_4(s)}$$

前向通道： $P_1 = G_4(s)$

单独回路： $L_1 = -G_2(s)G_4(s)$, $L_2 = -G_3(s)G_4(s)$

特征式： $\Delta = 1 + G_2(s)G_4(s) + G_3(s)G_4(s)$, $\Delta_1 = 1$

$$Y(s)/N(s) = \frac{1}{\Delta} P_1 \Delta_1 = \frac{G_4(s)}{1 + G_2(s)G_4(s) + G_3(s)G_4(s)}$$